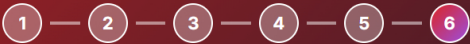


Learning Week: Day 1 → Day 6

A complete guide to understanding chains, agents, memory, orchestration, and building a real retail data pipeline



Day 1: Chains

Day 2: Templates

Day 3: Memory & Tools

Day 4: LangGraph

Day 5: Orchestrator

Day 6: The Project

Day 5: Orchestrator Agents

Multi-Agent Systems & Routing

What is an Orchestrator Agent?

An **orchestrator agent** receives a task, figures out what needs to happen, and routes it to the right specialized agent. It's like a manager — it doesn't do the work itself, it decides **who** should do it.

ANALOGY:

```
Customer Task: "I need a report on Q3 sales"
├ Orchestrator reads the task
├ Thinks: "This is a sales question"
├ Routes to → Sales Analysis Agent
└ Collects result & returns to user
```

Multi-Agent Systems

Instead of one giant agent that can do everything, you have multiple specialized agents:

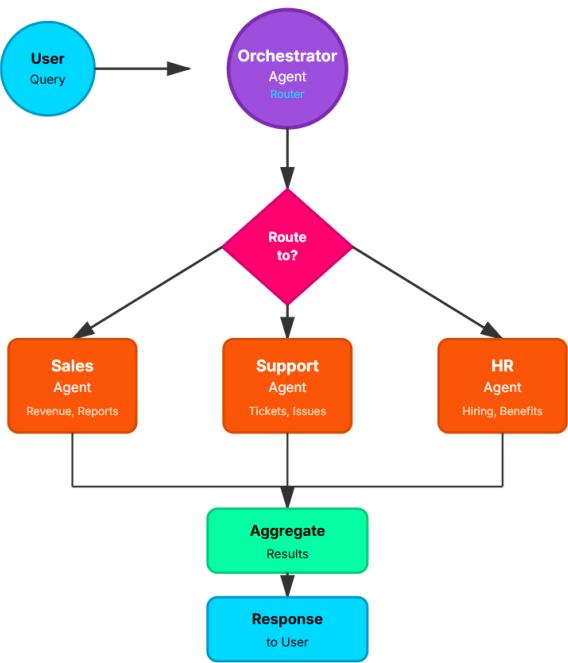
Sales Agent: Handles sales queries, revenue data

Support Agent: Handles customer support tickets

HR Agent: Handles hiring, benefits questions

Orchestrator: Decides which agent handles each task

Orchestrator Agent Flow



Key Points

- ✓ Orchestrator reads task
- ✓ Determines best-fit agent
- ✓ Routes task to agent(s)
- ✓ Collects results from all
- ✓ Aggregates into final reply

Why This Matters

Each agent specializes in one domain — more efficient, accurate, and easy to debug.

- Orchestrator
- Specialist Agent
- Aggregate / Output

Real Example

User Query: "What's my Q3 revenue and do I have any pending support tickets?"

1. Orchestrator receives the query
2. Recognizes TWO subtasks: sales question + support question
3. Routes to Sales Agent → gets revenue report
4. Routes to Support Agent → gets pending tickets
5. Combines both results
6. Sends to user: "Your Q3 revenue was \$500K. You have 3 pending tickets..."

Key Concepts

Orchestrator/Router Pattern: Central agent that routes tasks

Multi-Agent System: Multiple specialized agents working together

Shared State: Agents can pass information to each other

Supervisor Agent: Manages and coordinates agents

Task Decomposition: Breaking complex queries into subtasks

Why Our Next Project Needs Orchestrator Agents

The mini-project (Day 6) receives a raw retail CSV and must decide: clean first, then analyse, then visualise. Instead of one monolithic script, we build 4 specialised agents coordinated by an orchestrator.

Orchestrator Agent: Receives the raw CSV, decides the processing order, routes to each agent in sequence

Clean Agent: Fixes missing values, wrong data types, removes duplicates

Analysis Agent: Generates total sales, best-sellers, regional & category breakdowns

Visualization Agent (Bonus): Turns the analysis into bar, line, and pie charts

This design is modular, scalable, and much easier to debug. See the full agent specs in [Day 6: The Project →](#)

Key Takeaway: Orchestrator agents turn messy, complex problems into organized workflows. They separate concerns (each agent handles one thing), make systems more maintainable, and are how real AI applications scale.

Free Resources to Dive Deeper

LangChain Docs

Official documentation with API reference and guides

LangChain Academy

Free official courses on LangChain fundamentals

LangGraph Docs

Guide to building graphs and agents

Multi-Agent Tutorial

Real example of orchestrator agents in action

🌟 Learning Week: LangChain + LangGraph Fundamentals

DevSynt AI Automation Internship · Mentor: Afnan Shoukat

Quiz coming next — make sure you understand these concepts!